

Clinical Handbook

Comprehensive Condition Guide from Master Markdown Source

Prepared as a single readable document for clinical learning

BURNS — ELITE CLINICAL TRAINING DATASET (FULL ENGINE)

This section summarizes the clinical approach to this condition, including presentation-based assessment, diagnostic direction, and treatment planning.

Critical Severity Classification:

LIFE-THREATENING Airway burns (soot, hoarseness) Large burns > 20% body surface area Shock Inhalation injury #### MODERATE Partial thickness burns 10–20% body surface #### MILD Small superficial burns --.

Key Decision Points:

Burn size (total body surface area %) Depth (superficial, partial, full thickness) Airway involvement Cause (flame, chemical, electrical) Location (face, hands, genitalia) Signs of shock --.

WHEEZE — ELITE CLINICAL TRAINING DATASET (FULL ENGINE)

This section summarizes the clinical approach to this condition, including presentation-based assessment, diagnostic direction, and treatment planning.

Critical Severity Classification:

LIFE-THREATENING WHEEZE Silent chest Oxygen saturation < 90% Unable to speak full sentences Exhaustion / altered consciousness #### SEVERE Marked breathlessness Use of accessory muscles oxygen saturation 90–94% #### MODERATE Talks in phrases Mild distress #### MILD Talking normally Minimal symptoms --.

Key Decision Points:

Severity (most important) Known asthma vs first episode Age (infant vs adult) Fever (infection trigger) Response to bronchodilator Signs of pneumonia (crackles, fever) Signs of obstruction (no response) --.

EPIGASTRIC PAIN — Doctor/Assistant Training Conversation Variations

This section summarizes the clinical approach to this condition, including presentation-based assessment, diagnostic direction, and treatment planning.

Purpose: Train The Assistant To Handle ****Epigastric Pain**** Naturally, Safely, And Condition-Specifically.:

Source logic used: Epigastric pain causes Upper gastrointestinal bleeding linkage Peptic ulcer disease Gastritis Pancreatitis Biliary disease nonsteroidal anti-inflammatory drug use Helicobacter pylori Pain radiation Vomiting patterns Haematemesis / melaena Severity and instability --.

Epigastric Pain Decision Points:

1. Emergency features Haematemesis Melaena Shock → Upper GI bleeding pathway #### 2. Pain radiation Radiates to back → pancreatitis #### 3. Medication history nonsteroidal anti-inflammatory drug use → ulcer risk #### 4. Infection Helicobacter pylori → ulcer disease #### 5. Severity Mild →

gastritis Severe → admit / investigate --.

Condition Agreement / Clarification

This section summarizes the clinical approach to this condition, including presentation-based assessment, diagnostic direction, and treatment planning.

21. Doctor Vague Request:

Help manage pain. Do you mean epigastric or upper abdominal pain? Yes. Is there bleeding or severe symptoms? --.

22. Doctor Wrong Condition:

Pain from trauma. That is trauma, not epigastric pain condition. --.

23. Doctor Says Ulcer:

Manage ulcer pain. Do you mean peptic ulcer disease? Yes. Start PPI and test for Helicobacter pylori. --.

CHEST TRAUMA — ADVANCED CLINICAL TRAINING DATASET (FULL ENGINE)

This section summarizes the clinical approach to this condition, including presentation-based assessment, diagnostic direction, and treatment planning.

Key Life-Threatening Conditions:

Tension pneumothorax Open pneumothorax Massive haemothorax Cardiac tamponade Flail chest --.

Key Decision Points:

1. Stability Hypotension, hypoxia, distress → unstable Stable → imaging allowed ### 2. Breathing issues Reduced breath sounds Tracheal deviation Chest wall movement Oxygen saturation ### 3. Circulation Shock Distended neck veins (tamponade) Bleeding --.

ABDOMINAL INJURY — ADVANCED CLINICAL TRAINING DATASET (FULL ENGINE)

This section summarizes the clinical approach to this condition, including presentation-based assessment, diagnostic direction, and treatment planning.

Key Decision Framework:

1. Stability Shock → immediate resuscitation + surgery Stable → imaging (focused assessment with sonography in trauma / computed tomography) ### 2. Mechanism Blunt trauma → hidden bleeding Penetrating → high surgical risk ### 3. Red Flags Abdominal distension Guarding / rigidity Hypotension Tachycardia Reduced urine output Drop in haemoglobin ### 4. Investigations focused assessment with sonography in trauma (bedside) computed tomography abdomen (stable only) Full haemogram Cross-match --.

Condition Agreement / Clarification Training

This section summarizes the clinical approach to this condition, including presentation-based assessment, diagnostic direction, and treatment planning.

91. Doctor Asks Vaguely:

Can you help me manage bleeding? Do you mean upper gastrointestinal bleeding, such as vomiting blood or passing black stool? Yes. Okay. Is the patient unstable or actively bleeding? --.

92. Doctor Says Ugib Abbreviation:

Manage UGIB. Do you mean upper gastrointestinal bleeding? Yes. Is the patient haemodynamically unstable, or are there signs such as shock, confusion, tachycardia, or active haematemesis? --.

93. Doctor Says “Vomiting Blood”:

Vomiting blood. Do you mean haematemesis from suspected upper gastrointestinal bleeding? Yes. Is there shock or haemodynamic instability? --.

94. Doctor Rejects Upper Gi Bleeding:

I want to manage bleeding. Do you mean upper gastrointestinal bleeding, such as haematemesis or melaena? No, per vaginal bleeding. Understood. For per vaginal bleeding, first clarify whether the patient is pregnant, postpartum, or confirmed not pregnant. --.

95. Doctor Says “Black Stool”:

Patient has black stool. Do you mean melaena, suggesting possible upper gastrointestinal bleeding? Yes. Is the patient dizzy, collapsed, tachycardic, hypotensive, or actively bleeding? --.

96. Doctor Says “Fresh Blood Stool”:

Fresh blood per rectum. Fresh rectal bleeding may be lower gastrointestinal bleeding, but if the patient is unstable it can still be brisk upper gastrointestinal bleeding. Is the patient in shock or haemodynamically unstable? --.

97. Doctor Wants Variceal Confirmation:

How do I manage oesophageal varices bleeding? Do you mean upper gastrointestinal bleeding in a patient with known or suspected liver disease or portal hypertension? Yes. Treat as suspected variceal bleeding: careful resuscitation, avoid over-transfusion, urgent endoscopic control, and immediate specialist involvement. --.

98. Doctor Wants Ulcer Confirmation:

How do I manage ulcer bleed? Do you mean non-variceal upper gastrointestinal bleeding, such as peptic ulcer bleeding? Yes. Is the patient unstable, actively bleeding, or does the patient have low haemoglobin? --.

99. Doctor Gives Wrong Category:

Patient has epigastric pain but no bleeding. Epigastric pain alone is not the same as upper gastrointestinal bleeding. Has the patient had haematemesis, melaena, coffee-ground vomiting, or collapse with suspected gastrointestinal bleeding? --.

100. Doctor Asks For Summary:

Summarize upper GI bleeding management. First stabilize if unstable: large-bore intravenous access, isotonic fluids, full haemogram, cross-match, transfusion planning, and urgent endoscopy. Then separate variceal from non-variceal bleeding. Liver disease or known varices suggests variceal bleeding and needs careful resuscitation, avoidance of over-transfusion, urgent endoscopic control, and specialist involvement. For non-variceal bleeding, manage transfusion based on haemoglobin and active bleeding, consider tranexamic acid only in selected cases, and treat *Helicobacter pylori* if positive. --.

HEAD INJURY — CLINICAL TRAINING DATASET (FULL ENGINE)

This section summarizes the clinical approach to this condition, including presentation-based assessment, diagnostic direction, and treatment planning.

Core Rules:

Do NOT ask irrelevant questions (e.g., pregnancy). Use provided data immediately. Prioritize severity (Glasgow Coma Scale, airway, hypoxia, hypotension). Ask only ONE decision-changing question. Never repeat questions. Do not ignore new information. Always escalate when Glasgow Coma Scale ≤ 8 . --.

Key Decision Points:

Glasgow Coma Scale (Glasgow Coma Scale) $\leq 8 \rightarrow$ severe head injury 9–12 $\rightarrow$ moderate 13–15 $\rightarrow$ mild Airway compromise Hypoxia Hypotension Seizures Vomiting / confusion / unequal pupils computed tomography availability --.

ELEVATED BLOOD PRESSURE IN PREGNANCY — Doctor/Assistant Training Conversation Variations

This section summarizes the clinical approach to this condition, including presentation-based assessment, diagnostic direction, and treatment planning.

Purpose: Train The Assistant To Handle ****Elevated Blood Pressure In Pregnancy** Naturally, Safely, And Condition-Specifically.:**

Source logic used: Elevated blood pressure at least 140/90 millimetres of mercury. Pregnancy before 20 weeks versus after 20 weeks. Severe hypertension at or above 160/110 millimetres of mercury. Target organ symptoms: severe headache, blurred vision, epigastric pain, oliguria, liver tenderness. Proteinuria. Full haemogram: normal, haemoglobin below 10 grams per decilitre, thrombocytopenia below 100. Liver function tests: normal or elevated transaminases. Urea, electrolytes, and creatinine: normal or deranged. Coagulation profile: normal or abnormal. Obstetric ultrasound, uric acid, peripheral blood film. Doppler ultrasound of renals, electrocardiogram, echocardiography, kidney-ureter-bladder ultrasound, 24-hour urine for vanillylmandelic acid and metanephrines, unconjugated catecholamines where chronic hypertension/secondary cause is considered. Supportive management: reassure, left lateral position, cautious fluids. Treatment: methyldopa 250 milligrams three times daily for non-severe/chronic or mild disease where indicated; intravenous hydralazine 5 milligrams slowly over 10 minutes for blood pressure above 160/110; magnesium sulfate loading 4 grams of 20 percent over 15 minutes followed by 1 gram hourly for 24 hours while monitoring respiratory rate, urine output, and patellar reflexes. --.

Key Clinical Decision Points:

1. Gestational age Below 20 weeks: consider chronic hypertension or secondary hypertension. At or beyond 20 weeks: consider gestational hypertension, mild pre-eclampsia, severe pre-eclampsia, or eclampsia. Postpartum: eclampsia/severe hypertension may still occur. ### 2. Blood pressure severity At least 140/90: elevated blood pressure. 140 to 179 systolic or 90 to 109 diastolic without target organ damage: non-severe range, still needs evaluation. 160/110 or higher: severe hypertension requiring urgent antihypertensive management. ### 3. Target organ symptoms Severe headache with blurred vision. Epigastric pain. Oliguria. Liver tenderness. Impaired consciousness. Pulmonary oedema. ### 4. Seizure New-onset seizure in pregnancy or postpartum is eclampsia until proven otherwise. ### 5. Laboratory interpretation Proteinuria supports pre-eclampsia. Elevated transaminases suggest severe disease. Deranged urea, electrolytes, and creatinine suggest renal impairment. Platelets below 100 suggest severe disease or HELLP-type concern. Abnormal coagulation

profile suggests severe disease and bleeding risk. Haemoglobin below 10 means anaemia that should be considered in management. #### 6. Investigations Full haemogram. Urinalysis or urine protein assessment. Urea, electrolytes, and creatinine. Liver function tests. Coagulation profile. Obstetric ultrasound. Uric acid. Peripheral blood film. Renal Doppler, electrocardiogram, echocardiography, kidney-ureter-bladder ultrasound, and 24-hour urine tests when chronic hypertension or secondary causes are considered. --.

HOTNESS OF BODY / FEVER — Doctor/Assistant Training Conversation Variations

This section summarizes the clinical approach to this condition, including presentation-based assessment, diagnostic direction, and treatment planning.

Purpose: Train The Assistant To Handle **Hotness Of Body / Fever Naturally, Safely, And Condition-Specifically.:**

Source logic used: Fever in children and adults. Fever less than 7 days. Danger signs first. Malaria risk and malaria antigen result. Hydration status. Respiratory symptoms. Urinary symptoms. Rash. Neck stiffness. Immunosuppression. Full haemogram. Urea, electrolytes, and creatinine. Random blood sugar. Stool results where gastroenteritis is suspected. Severe malaria indicators. Convulsions, hypoglycaemia, dehydration, anaemia, sepsis, meningitis, and gastroenteritis-related branches. --.

Fever Decision Points:

1. Severity / danger signs Danger signs include: impaired consciousness convulsions or repeated seizures shock severe dehydration severe anaemia acidosis respiratory distress lethargy inability to drink neck stiffness abnormal posturing or hallucinations #### 2. Age group Less than 60 days: high-risk fever, lower threshold for admission and sepsis workup. More than 60 days: assess danger signs, hydration, malaria risk, respiratory symptoms, urinary symptoms, rash, neck stiffness, and immunosuppression. #### 3. Malaria result Malaria positive and stable with oral intake: outpatient antimalarial treatment may be appropriate. Malaria positive with danger signs, anaemia, acidosis, shock, or renal derangement: severe malaria concern; admit and start parenteral artesunate. Malaria negative: look for other sources such as viral illness, bacterial infection, urinary infection, pneumonia, meningitis, gastroenteritis, or sepsis. #### 4. Full haemogram Normal full haemogram with malaria negative: viral or atypical infection more likely if stable. Granulocytosis: bacterial infection, sepsis, gastroenteritis, or inflammatory focus. Lymphocytosis: viral infection or atypical illness. Haemoglobin below 10: anaemia; if severe or unstable, admit. Platelets very low: consider severe infection, malaria complication, bleeding risk, or haematological cause. #### 5. Urea, electrolytes, and creatinine Normal: less concern for dehydration-related acute kidney injury. Deranged: dehydration, acute kidney injury, severe malaria, sepsis, or gastroenteritis complication. #### 6. Random blood sugar Low glucose: treat immediately with dextrose and admit. Normal glucose: continue evaluating fever source. #### 7. Stool / gastroenteritis features Vomiting, diarrhoea, dehydration, abnormal stool: assess hydration and consider gastroenteritis pathway. Unable to retain oral fluids: admit. #### 8. Respiratory features Cough, fast breathing, chest signs, low oxygen: assess for pneumonia or respiratory infection. #### 9. Neurological features Convulsions, neck stiffness, altered consciousness, abnormal posturing, hallucinations: admit; consider meningitis, encephalopathy, severe malaria, hypoglycaemia, trauma, or intracranial pathology. --.

Condition Agreement / Clarification Training

This section summarizes the clinical approach to this condition, including presentation-based assessment, diagnostic direction, and treatment planning.

71. Doctor Asks Vaguely:

Can you help me manage hotness? Do you mean hotness of body, meaning fever? Yes. Okay. Are there danger signs such as confusion, convulsions, shock, severe dehydration, neck stiffness, or breathing difficulty? --.

72. Doctor Gives Wrong Condition:

Hotness of body from burns. Do you mean fever, or a burn injury? Burn injury. Understood. That is a burns presentation, not fever. For burns, the important details are percentage of body surface area, burn depth, and airway risk such as facial burns, hoarseness, stridor, or soot in sputum. --.

73. Doctor Says “Malaria Fever”:

I want to manage malaria fever. Do you mean a patient with fever and a positive malaria test? Yes. Is the patient stable with oral intake, or are there severe features such as impaired consciousness, repeated seizures, shock, severe anaemia, acidosis, or renal derangement? --.

74. Doctor Says “Infection Fever”:

I want to manage infection fever. Do you mean fever where bacterial infection or sepsis is suspected? Yes. Is the patient toxic-looking, hypotensive, confused, tachypnoeic, or unable to drink? --.

75. Doctor Rejects Fever Match:

How do I manage hot body? Do you mean fever? No, I mean hyperthermia after heat exposure. Understood. That is different from ordinary fever. Please share the temperature, mental status, hydration status, and whether there was heat exposure or exertion. --.

PV BLEEDING — Doctor/Assistant Training Conversation Variations

This section summarizes the clinical approach to this condition, including presentation-based assessment, diagnostic direction, and treatment planning.

Purpose: Train The Assistant To Handle **Pv Bleeding Naturally, Safely, And Condition-Specifically.:**

Source logic used: the PV bleeding pathways are separated into non-pregnant abnormal uterine bleeding, pregnancy below 20 weeks, pregnancy at or beyond 20 weeks, postpartum haemorrhage, sepsis-associated bleeding, and coagulopathy-related bleeding. The assistant must first identify the relevant subgroup, then ask only pathway-specific questions. --.

Condition Agreement / Clarification Training

This section summarizes the clinical approach to this condition, including presentation-based assessment, diagnostic direction, and treatment planning.

Purpose: Sometimes The Doctor Does Not Start With A Case. They May Ask Generally: “I Want To Manage Bleeding,” “I Want Pv Bleeding,” Or “How Do I Manage This Condition?”:

In that situation, the assistant should confirm the closest condition name naturally before continuing. Rules: If the doctor gives a vague condition name, look for the closest condition in the clinical database. Confirm naturally: “Do you mean a patient with...?” If the doctor says yes, continue with that condition’s normal workflow. If the doctor says no, ask what exact condition or presentation they want to manage. Do not ask pregnancy, gestational age, shock, or ultrasound questions until the condition is agreed. Once the condition is agreed, continue with condition-

specific questions only. Keep the tone conversational. --.

41. Doctor Asks Vaguely For “Bleeding”:

I want to manage bleeding. Do you mean a patient with per vaginal bleeding? Yes. Okay. For per vaginal bleeding, the first important distinction is whether this is pregnancy-related. Is the patient pregnant, postpartum, or confirmed not pregnant? --.

42. Doctor Asks For “Pv Bleeding”:

How do I manage PV bleeding? Do you mean per vaginal bleeding? Yes. Good. The next step is to identify the subgroup because management differs for non-pregnant bleeding, early pregnancy bleeding, late pregnancy bleeding, and postpartum bleeding. Is the patient pregnant, postpartum, or confirmed not pregnant? --.

43. Doctor Says No To Suggested Condition:

I want to manage bleeding. Do you mean a patient with per vaginal bleeding? No. What exact bleeding presentation do you want to manage — for example upper gastrointestinal bleeding, postpartum bleeding, trauma-related bleeding, or another condition? Upper gastrointestinal bleeding. Understood. That is different from per vaginal bleeding. Please share whether the patient has haematemesis, melaena, haematochezia, shock, or known liver disease so we can choose the correct approach. --.

44. Doctor Gives A Misspelled Condition:

How do I manage pv bleding? Do you mean per vaginal bleeding? Yes. Okay. Is the patient pregnant, postpartum, or confirmed not pregnant? --.

45. Doctor Gives A Broad Obstetric Complaint:

I want to manage bleeding in pregnancy. Do you mean per vaginal bleeding during pregnancy? Yes. The next important detail is gestational age because early pregnancy bleeding and bleeding after 20 weeks are managed differently. Is the pregnancy below 20 weeks or at least 20 weeks? --.

46. Doctor Gives Enough Condition Detail; No Confirmation Needed:

Pregnant 30 weeks with painless per vaginal bleeding. Painless bleeding at 30 weeks raises concern for placenta previa until excluded. Avoid vaginal examination and arrange urgent obstetric ultrasound. Has placenta previa been confirmed? --.

47. Doctor Asks For A Named Subgroup:

How do I manage placenta previa? Do you mean a pregnant patient at or beyond 20 weeks with suspected or confirmed placenta previa? Yes, confirmed placenta previa. With confirmed placenta previa, management depends on whether bleeding is active or recurrent. Is bleeding active or recurrent? --.

48. Doctor Rejects The Closest Match:

How do I manage placenta previa? Do you mean a pregnant patient at or beyond 20 weeks with suspected or confirmed placenta previa? No, I mean placental abruption. Understood. Placental abruption is more likely when bleeding is associated with uterine tenderness, abnormal fetal heart rate, rising maternal pulse, or falling blood pressure. Is there uterine tenderness or abnormal fetal heart rate? --.

49. Doctor Asks “What Condition Is This?”:

Patient has vaginal bleeding and lower abdominal pain. What condition is this? This may still be per vaginal bleeding, but the likely cause depends heavily on pregnancy status and gestational age. Is the patient pregnant? Yes, 9 weeks. At 9 weeks, this is early pregnancy bleeding. The key concern is to exclude ectopic pregnancy and assess for miscarriage-related causes. Arrange obstetric ultrasound

and serum human chorionic gonadotropin trend. Is she haemodynamically stable? --.

50. Doctor Wants Model To Confirm Database Condition Name:

I want to manage abnormal uterine bleeding. Do you mean abnormal uterine bleeding in a non-pregnant patient? Yes. Okay. Is the patient in shock, or is she haemodynamically stable? --.

51. Doctor Says Condition Only, No Patient Details:

Placenta previa. Do you mean a pregnant patient at or beyond 20 weeks with suspected or confirmed placenta previa? Yes. Has placenta previa been confirmed on ultrasound, and is bleeding active or recurrent? --.

52. Doctor Says “Post Delivery Bleeding”:

How do I manage bleeding after delivery? Do you mean postpartum haemorrhage? Yes. Okay. The likely cause matters. Is the uterus soft or boggy? --.

53. Doctor Says “Young Woman Bleeding For Years”:

Young woman with heavy menses for years and gum bleeding. This sounds more like bleeding related to an underlying coagulation disorder, such as von Willebrand disease. Do you want to manage this as coagulopathy-related per vaginal bleeding? Yes. Measure von Willebrand factor antigen, ristocetin cofactor activity, and factor VIII level. Refer to haematology before starting desmopressin or von Willebrand factor concentrate. --.

54. Doctor Gives A Condition Outside Pv Bleeding:

How do I manage head injury? That is not a per vaginal bleeding presentation. Do you want to continue with head injury management instead? Yes. Okay. For head injury, the key severity marker is the Glasgow Coma Scale score. What is the Glasgow Coma Scale score? --.

55. Doctor Gives Unclear Phrase “Bleeding Below”:

Patient has bleeding from below. Do you mean vaginal bleeding, rectal bleeding, or bleeding from the urinary tract? Vaginal bleeding. Okay, this is per vaginal bleeding. Is the patient pregnant, postpartum, or confirmed not pregnant? --.

56. Doctor Confirms Wrong Condition But Gives Contradictory Clinical Details:

I want to manage abnormal uterine bleeding. Do you mean abnormal uterine bleeding in a non-pregnant patient? Yes, she is 24 weeks pregnant. Since she is 24 weeks pregnant, this is not non-pregnant abnormal uterine bleeding. This is bleeding at or beyond 20 weeks of pregnancy. Is the bleeding painless, or is there uterine tenderness or abnormal fetal heart rate? --.

57. Doctor Gives Partial Subgroup: “Abortion”:

How do I manage abortion with bleeding? Do you mean bleeding in pregnancy below 20 weeks, such as threatened, incomplete, inevitable, missed, septic, or complete abortion? Yes, 12 weeks. At 12 weeks, manage as early pregnancy bleeding until ectopic pregnancy is excluded. Arrange obstetric ultrasound and serum human chorionic gonadotropin trend. Is she in shock? --.

58. Doctor Asks For “Septic Abortion”:

How do I manage septic abortion? Do you mean early pregnancy bleeding with infection features such as fever, uterine tenderness, offensive discharge, or granulocytosis? Yes, fever and offensive discharge. That clinical picture strongly suggests septic abortion. Start intravenous ceftriaxone and metronidazole immediately while arranging evacuation under gynaecological care. Assess haemodynamic status and send complete blood count, renal function, electrolytes, blood group, and cross-match. --.

59. Doctor Asks Without Case, Assistant Asks Condition:

Can you help me with management? Yes. What condition or presenting complaint do you want to manage? PV bleeding. Do you mean per vaginal bleeding? Yes. Is the patient pregnant, postpartum, or confirmed not pregnant? --.

60. Doctor Wants Differential Rather Than Management:

Patient has vaginal bleeding. What could it be? The main possibilities depend on pregnancy and timing. If not pregnant, this may be abnormal uterine bleeding or bleeding related to coagulopathy or infection. If pregnant below 20 weeks, consider miscarriage-related causes and exclude ectopic pregnancy. If pregnant at or beyond 20 weeks, consider placenta previa or placental abruption. If postpartum, consider postpartum haemorrhage from uterine atony, trauma, or retained placental tissue. Is the patient pregnant, postpartum, or confirmed not pregnant?

End of Handbook